

1 FINAL-STATE STABILITY DOES NOT GUARANTEE A SAFE TRANSITION

SG retirement changes a portfolio one unit at a time. Stability belongs to the whole portfolio, so a stable final state cannot certify every implementation order.

$$\alpha_{\perp}(S, \theta) = \max \operatorname{Re} \lambda(A_{\perp}(S, \theta))$$
$$\mathcal{H}(\theta) = \operatorname{Min}_{\subseteq} \{S : \alpha_{\perp}(S, \theta) \geq 0\}$$

Each hyperedge is an inclusion-minimal unstable replacement coalition.

synchronous generators → retire and replace in stages → IBR → inverter-based resources

A stable target can still be unreachable without crossing an unstable portfolio.

2 15/15 SMALLER PORTFOLIOS ARE STABLE; THE COMPLETE PORTFOLIO IS NOT

IEEE-39 one-line diagram

15/15 proper subsets stable

$H_4 = \{30, 33, 35, 37\}$
canonical, not universal
P4 control settings
 $g = 0.03625 \quad k = 1.425$
 $t = 1.5 \quad h = 1$
 $\alpha_{\perp} = +0.127 \text{ s}^{-1}$
 $f = 0.6223 \text{ Hz}$
Displaced: 2096.6 MW
Converter: 4270.7 MVA

Hasse lattice ($n = 4$)

$\{30, 33, 35, 37\}$ (unstable)

H_4 is a minimal instability hyperedge.

• synchronous generator
× highlighted IBR candidate

3 STABLE TARGET $\neq$ SAFE PATH

Three levels of safety and how they relate:

A final-state stable
 $\alpha_{\perp}(S, \theta) < 0$

B some safe order
 $\exists$ all prefixes stable

C every order safe
 $\forall$ orders safe

$C \Rightarrow B \Rightarrow A$
The converse does not hold.

$C(T) \Leftrightarrow e \not\subseteq T \quad \forall e \in \mathcal{H}(\theta)$

Why common screens fall short

Screen	Configuration	Blocker
SCR / gSCR	✓ local	×
Modal	✓ local	×
Additive	✓ approximate	×
Pairwise	✓ approximate	incomplete
Full eigensolve	✓ per case	enumerate all
$\mathcal{H}(\theta)$	✓ exact	explicit set

9-candidate safe-set census: PENDING.

15/15 smaller portfolios stable
all strict subsets

$H_4 = \{30, 33, 35, 37\}$
canonical witness
not the universal blocker

$+0.127 \text{ s}^{-1}$
critical growth rate
canonical portfolio

0.6223 Hz
critical frequency
nominal IEEE-39 case

4 THE DEVICES REMAIN REGULAR; THE NETWORK CLOSES THE INSTABILITY

Exact factorization

$$T_S(s) = T_0(s) + \sum_{i \in S} E_i \Delta Y_i(s) E_i^T$$
$$\det P_S(s) = h_S(s) \det T_0(s) \cdot \det(I + M_{SS}(s))$$
$$\det(I + M_{SS}) = \prod_{i \in S} \det(I + M_{ii}) \cdot \det(I + Q_{SS})$$
$$\det(I + Q_{SS}) = \sum_{T \subseteq S} \mu_T$$

Devices to collective closure

LOCAL DEVICE FACTORS
 $\prod_{i \in S} \det(I + M_{ii})$
each device remains regular

COLLECTIVE NETWORK CLOSURE
 $\det(I + Q_{SS})$
can close at zero

network-mediated collective return
The local factors stay finite while the network return closes.

Boundary behavior

$g^* = 0.207681; \alpha_{\perp} = 0; f = 0.7064 \text{ Hz}$

$\sigma_{\min}$

feedback gain g

• local × Q_H closure ♦ g^*

Every member is necessary in context
 H_4 is minimal; all proper subsets remain stable.

The devices remain locally regular.
Instability appears when the network-mediated collective return closes.
Strict M1 reduction: BLOCKED.
 $r = 2.34 \times 10^{-11}$; gate 1.00×10^{-12} .

5 THE SAME GRID CAN HAVE DIFFERENT MINIMAL BLOCKERS

IEEE-39 V4 policy atlas

Distinct hypergraphs
30 36 16
F7A F7B F7C

Blocker-cardinality sequence
 $4 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow \emptyset$
One grid, different policy, different blocker structure.
Previously audited atlas; current policy campaign status: CONDITIONAL.

Q/V gain g (horizontal); AVR gain k (vertical).
0 1 2 3 4 unsampled blocker order $|H_{\min}|$.

6 H_4 IS THE WITNESS — NOT THE UNIVERSAL BLOCKER

Frozen synthetic envelope (1000 scenario IDs)

967 valid (96.7%)

33 infeasible (3.3%) Scenario totals await review.

190/967
19.6%
 H_4 minimal

555/967
57.4%
any V_4 collective blocker

365/967
37.7%
alternative witnesses

PENDING USER REVIEW
Descriptive synthetic ensemble; IDs are not a probability sample. H_4 is minimal in the audited event.

7 A FIX THAT REPAIRS THE CANONICAL CASE IS NOT A UNIVERSAL CURE

Control retuning: $g = 0.03625 \rightarrow 0.25$

594/967 61.4%
 $\alpha_{\perp}$ improves
• improved × deteriorated

186/967 19.2%
canonical H_4 rescued

373/967 38.6%
stability deteriorates

Canonical phasor-DAE response (0–30 s)

$\Delta \omega$ (p.u.)

time (s)

— H_4 , original
— H_4 , retuned
... proper triple

TDS 56/60; 4 zero-Hz; 30 failed

Envelope: PENDING

PENDING USER REVIEW
Support: 106.0 MVA, conditional

8 AVOIDING EVERY HYPEREDGE PROTECTS THE WHOLE IMPLEMENTATION PATH

Path-constrained planning formulation

$$\max_S \sum_{i \in S} P_i \quad \text{s.t. } e \not\subseteq S, \quad \forall e \in \mathcal{H}(\theta)$$

Apply the same constraint to every intermediate prefix of the implementation path.

Unsafe implementation path
Schematic: the path reaches the canonical hyperedge.

Safe implementation path
Schematic: every prefix avoids every blocker.

Reject the route when a prefix contains H_4 .

100/100 cross-code

56/60 finite TDS agreement

P_{∞} PENDING safe-set census pending

9 EVIDENCE IS MEASURED, SCOPED, AND READY TO AUDIT

100/100 same-model cross-code

56/60 finite TDS sign agreement

15/15 proper subsets stable

30/36/16 distinct policy blockers

594/967 improved; review pending

190/967 H_4 minimal; review pending

Scope: phasor-domain DAE; stationary, re-equilibrated portfolios; no physical switching transient, EMT, current-limit, or DC-link claim; blocker identity depends on model and control policy.

REFERENCES

[1] Kundur, *Power System Stability*, 1994.
[2] Kundur et al., *IEEE Trans. Power Syst.* 19, 2004.
[3] Hatziaargyriou et al., *IEEE Trans. Power Syst.* 36, 2021.
[4] Dörfler et al., *PNAS* 110, 2013.
[5] Milano, *Power System Modelling*, 2010.

A SAFE GENERATION TRANSITION IS NOT DEFINED BY THE NEXT DEVICE.

It is defined by which replacement coalitions can coexist, under which control policy, along the entire implementation path.

CONTACT

Jorge Luis Mayorga Taborda
j.l.mayorga@uniandes.edu.co
Universidad de los Andes
Bogotá, Colombia

Code + data

IEEE IAS Annual Meeting 2026

Vancouver, Canada • October 4–8, 2026